

The Bridge Certificate: A Line-Level Audit Instrument for Inter-Theoretic Transfer Arguments, with Four Applications to the *abc* Literature

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Abstract

A recurring shape of argument in arithmetic geometry moves a numerical readout out of one construction into a container it shares with another, and then reads an inequality off the result. Disputes about such steps have proved unusually resistant to the ordinary instruments of mathematical review: the parties agree on almost every formula and disagree about whether a particular passage *is* an argument. This paper proposes an instrument for exactly that situation, and reports what it returned in four applications. The instrument — a *bridge certificate* — fixes seven obligations of the proposed certificate for passages from a qualitative linkage to a numerical membership statement, arranges them as a nine-row ledger to be filled line by line against the wording of a primary text, and constrains how findings may be recorded: verdicts are about certificate form, never about mathematical truth; every finding carries a verbatim quotation with a line anchor; gaps are typed rather than graded; every appeal to standard literature is entered in an import ledger; and every claim of the form “this is not in the text” is reported together with the search terms used. Four applications are given, and they do four different things. Against the proof of one corollary of inter-universal Teichmüller theory the ledger *locates* an unmet obligation: the decisive membership is stated to follow “formally” from a pair of declared properties rather than being derived. Against the follow-up paper that the same text names as its quantification site, the ledger *closes off* that site: the computation there is explicit and, at every node it names, self-contained — this paper did not recompute it — but it consumes the disputed inequality as an input rather than producing it. Against an independent alternative program in a different theory the ledger serves as a *comparative discrimination case* and returns a differently shaped gap at a different address. Against two downstream applications it records *inheritance*: the disputed step is imported in a single sentence and never discussed. In the same period an international formalization project, which does not cite this work, isolated a compatibility statement at the same passage; we map its objects onto the certificate’s obligations by their proposed roles. This is not a proved implication from the report’s compatibility equation to the certificate’s typed order condition. Nothing in this paper is a claim that any argument examined is wrong, and nothing in it is a statement about the *abc* conjecture.

Scope and claim boundary.

- This paper proves *nothing* about the *abc* conjecture. It contains no proof and no disproof of it, and no statement about whether it is provable by the methods discussed.
- It contains *no error claim*. A ledger row returned as “not delivered” records that a stated obligation is not discharged *in the wording examined*. That is a statement about the form of a certificate, not about the truth of a theorem. “Not delivered” is not “refuted”.
- It *verifies no bound*. No proof line was followed step by step, no estimate was recomputed, no constant was checked. Where a computation is reported as carried out in a source, that is a finding *that* it is carried out and *where*, not a confirmation that it is correct.
- It takes no side in any existing dispute about the arguments examined. The known objections enter this paper only as the definition of a negative control, never as evidence.
- **No authority argument.** Neither an author’s own assertion that a step is sufficient, nor a critic’s assertion that it is not, counts as evidence here. Self-assessments found in the examined texts are objects of examination, not support.
- **Textual basis** are `pdftotext` extracts of the source PDFs. All line references (“extract 1. NNNN”) are finding aids into those extracts and are *not* pagination of the published documents. Unlike in a purely prose-level audit, one verdict of Section 3 does depend on symbols lost in extraction; the reconstruction is disclosed in Section 3.2 and the rule governing such cases is part of the instrument (Section 2.8).
- Bibliographic entries are provisional and carry their provenance; three of them are deliberately reduced to what could be verified. See Section 8, item 9.

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Changes in version 0.3

This corrective version distinguishes the prospective recipe from the partial historical applications, corrects the Case-C synopsis reference to ATS IV v2 § 6.10/Theorem 6.10.1, and restores the LANA equation’s notation $\eta^q = \eta_S^{\text{anab}}$ (9-1). The LANA crosswalk is a proposed role correspondence: neither a volume quotient nor equality of abstract maps by itself proves the certificate’s typed half-line condition. The required object, evaluation and bound remain open. The seven obligations define the proposed form contract; their universal necessity for every possible mathematical proof is not established. Case D concerns the Zhou downstream imports, not a completed C7 positive control or a demonstrated dependency cycle. Case C does not prove a mathematical contradiction in ATS. The previously prepared numbered table captions and two synopsis tables are retained. English and German texts incorporate the same corrections. No new mathematical result or independent method validation is claimed.

Changes in version 0.2

Version 0.1 of this paper appeared before the methods paper of the series [13] was finished. That paper hardened the sixth contract component: a common numerical evaluation in the container no longer discharges it by itself; the compatibility identities the claim needs — in particular $\nu_C \circ i_q = \nu_q$ on the stated source domain — are mandatory, and a stipulated invariance is not a proof of one. Version 0.2 adopts that reading, and adopting it changes two verdicts that version 0.1 recorded. A published coding should not be revised silently, so both are named here.

- **Case A, obligation 6: discharged → does not deliver.** Version 0.1 recorded obligation 6 as met “via its first alternative”, because a numerical evaluation exists in the container, and stated that “the gap sits at obligation 7, not at obligation 6”. Under the compatibility reading the same evidence yields the opposite coding, and the evidence is the paper’s own: the “searched, not found” block of Section 3 already recorded that the identity $\nu_C \circ i_q = \nu_q$ across the container identification is not in the text, and that the one compatibility lemma which does exist lies on another axis. No source was re-examined for this change and no new evidence was gathered; what changed is the rule the existing evidence is read against.
- **Case C, obligation 6: discharged → partly.** Version 0.1 recorded obligation 6 as met, again via the first alternative. The evaluation there is genuinely numerical, but the compatibility holds only between the sums and is place-wise disclaimed by the source itself — in a passage version 0.1 already quoted, one paragraph further on.

Three consequences are carried through the text rather than confined to this note: obligation 6 receives a ledger row of its own (the R6a / R6b split of Section 2.4); the application recipe of Section 2.11 is corrected at step 4, so that a re-use of it does not reproduce the superseded reading; and the summary statements which flanked the Case A finding with a discharged obligation 6 — in the results table, in the role table of Section 7 and in the conclusion — are brought into line.

What did not change. The central finding is untouched: obligation 7 is not discharged at the substep examined, and the membership is stated to follow “formally” from a property pair. The abstract of version 0.1 did not carry the obligation-6 claim and is unchanged in substance. Cases B and D keep their verdicts, and no case was re-audited.

The remaining changes are terminological , and align this paper with the finished methods paper. The instrument is given a version label and its unprefix row labels are scoped (Section 2.3). Row R8 is identified as a triggered extension rather than a core row, which corrects the denominator without changing its coding. And Case C is renamed from “comparison control” to *comparative discrimination case*, the term Part 2 uses — a change of label only, since this paper already recorded that a genuine positive control was not among the cases examined.

1 Introduction

1.1 A disagreement that ordinary review does not resolve

There is a class of mathematical disagreement in which the usual repair mechanism fails. It is not that one side has made an arithmetic slip and the other has found it. It is not that a lemma is false and a counterexample settles the matter. It is that a passage of text is read by its author as an argument and by a reader as a restatement, and that both readings survive contact with the formulas, because the formulas are not in dispute.

The passage at the center of this paper has that character. In the third volume of inter-universal Teichmüller theory, the proof of a corollary reaches a point at which two quantities — a numerical readout attached to one object, and a half-line of real numbers determined by another — are brought into a single evaluation, and an inclusion of the first in the second is then asserted. Around that inclusion an entire arithmetic consequence is organized. Whether the passage establishes the inclusion or presupposes it has been argued about for years without convergence.

Ordinary review prose is a poor instrument here, for a structural reason. Prose review answers the question “is this correct?”. When the disagreement is about whether a given sequence of sentences constitutes a derivation at all, that question has no stable meaning: each side answers it about a different object. Worse, prose review permits — indeed invites — a move that makes the disagreement unresolvable in principle, namely the appeal to authority. “The author says the construction suffices” and “the critics say it does not” are both statements about people. Once they enter, the text has stopped being the subject.

1.2 What an audit instrument does instead

The alternative pursued here is to stop asking whether the argument is correct and start asking a question that a text can answer on its own: *which obligations does a passage of this kind have to discharge, and which of them does this particular wording discharge?*

That reformulation buys four things.

1. **It is decidable from the text.** Whether a document contains a lemma asserting a given implication is a question about the document. Two careful readers can disagree about whether a proof is convincing; they will rarely disagree about whether a sentence is present.
2. **It localizes.** A verdict is attached to a numbered obligation and, through it, to a line range of a specific text. “The argument has a gap” becomes “obligation 7 is not discharged at this passage, and here is the sentence that stands in its place”.
3. **It is symmetric.** The same instrument, unchanged, is applied to a proposal and to its competitor. A method that returns “fails” for one text and has no procedure for returning anything about another is not an instrument but a verdict looking for a subject.
4. **It bounds its own claims.** Because the question is about form, the answer cannot silently grow into a claim about truth. An unmet obligation is a missing certificate component, not a refutation — and the instrument is built so that the distinction cannot be lost in the reporting.

The price is equally clear and is stated here once, because everything in this paper inherits it: an instrument of this kind can never establish that an argument is sound. Every obligation may be discharged and the argument may still be wrong for a reason the instrument does not look for. It can only establish that a specific, named requirement is or is not met by a specific, quoted wording.

1.3 The instrument, in one paragraph

A *bridge certificate*, written Br_ξ , is a list of seven obligations attached to an argumentative step of a particular shape: a step that carries a numerical readout out of one construction, into a container shared with a second construction, and then reads an order relation off the result. The obligations demand, in outline, that both sides carry explicit numerical readouts, that the shared container come with typed embeddings, that the identification between the sides be a permitted identification rather than a coincidence of names, that the indeterminacy budget be split into absorbed and unabsorbed parts, that a numerical evaluation exist in the container, and — the last and decisive one — that the order relation be *derived* from the permitted identification rather than asserted alongside it. The obligations are operationalized as a ledger of nine rows, each a question with a recorded expectation and a recorded risk, to be filled against a primary text. Around the ledger sit five disciplines that make its findings auditable: a verdict vocabulary that separates form from truth, a typology that classifies gaps instead of grading them, an import ledger for every appeal to standard literature, a rule for reporting what was searched for and not found, and a rule for disclosing reconstructions where the textual basis is imperfect. Section 2 states all of this in full.

1.4 Sources, labels and the anchor convention

Seven documents are used, in the form of text extracts of their PDFs: six carry the four applications, and the seventh is the interim report examined in Section 7. The short labels of the left column are used throughout; every source citation in this paper carries one, because the extracts have independent line numbering and a bare line number would be ambiguous (see Table 1).

Label	Source	Role in this paper
IUT-III	[1]	Case A: the corollary and its proof
IUT-IV	[2]	Case B: the declared quantification site
ATS-III	[5]	Case C: construction and volume theory
ATS-IV	[6]	Case C: the <i>abc</i> conclusion and its bridge
Zhou-I	[7]	Case D: first downstream application
Zhou-II	[8]	Case D: second downstream application
LANA	[9]	Section 7: independent formalization

Table 1: Primary source corpus and analytical roles in the four case audits.

Quotations are verbatim from the extracts; mathematical symbols have been typeset, ellipses are marked, and every reconstruction of a symbol lost in extraction is disclosed where it occurs. Emphasis inside quotations is ours unless stated otherwise.

Language of quotations. A German version of this paper exists, and it keeps every quotation in the English original. That is not an omission of the translation but a consequence of the subject matter: here the wording *is* the evidence. Whether a text writes “follows formally” or proves a theorem is precisely the question at issue, and translating such a passage would replace the finding by its rendering. Only the surrounding assessment is translated.

1.5 What follows

Section 2 states the instrument in full — all seven obligations and all nine rows — and Section 2.11 gives the application recipe, so that it can be executed against a transfer argument other than the four examined here. Sections 3–6 report the four applications. Section 7

states what they jointly show — and, at equal length, what they do not. Section 8 lists the limitations, Section 9 concludes.

Place in the series. This is Part 3 of the *abc* Gap Series. Part 1 is the published forensic study of the same passage [12], on which Section 3 builds and from which it is delimited there. The certificate stated here is one instance of a broader gate methodology — the instance for numerical membership — and that broader methodology has since appeared as Part 2 of the series [13]. Version 0.1 of this paper assumed nothing from it, because it did not yet exist. This version takes one thing from it and says where: the hardened reading of obligation 6 (Section 2.2), which is why two verdicts of the first version are recoded here — see “Changes in version 0.2” above. What this paper contributes is the instance and the four applications reported below, and those applications fill the ledger only in part: Case A reports four of its nine rows, as Sections 3 and 7 state.

2 The bridge certificate

2.1 The question

The instrument addresses steps of the following shape. Two constructions produce two objects. The first carries a numerical readout ν_q — in the cases below, a normalized logarithmic volume. The second determines a target set, in the cases below a closed real half-line H_Θ . The two objects are then exhibited inside a common container, related by data that is variously described as a linkage, an identification, a transport or a correspondence, and an assertion of the form

$$\nu_q(\text{first object}) \in H_\Theta$$

is made. From that membership an inequality follows immediately, and the inequality is the point of the whole construction.

The question the certificate asks is:

When may a qualitative package — two objects, shared labels, an algorithmic transport, a common container — be read as a numerical membership statement in a concrete real half-line?

This is deliberately narrower than “is the argument correct?” and deliberately wider than “is this formula right?”. It is the question that the disagreement is actually about.

2.2 The seven obligations

In the proposed specification, an admissible certificate Br_ξ contains at least the following. These are obligations of this chosen form contract, not proved necessary conditions for every possible mathematical proof.

Obligation 1 (*q*-side object with readout). An object Obj_q together with a concrete log-volumetric readout ν_q .

Obligation 2 (Θ -side object with target). An object Obj_Θ together with the target half-line $H_\Theta = \mathbb{R}_{\leq -|\log(\Theta)|}$.

Obligation 3 (Common container with typed embeddings). A container C together with explicitly typed embeddings i_q, i_Θ .

Obligation 4 (Linkage data). Label and linkage data L which state not merely that the two sides carry the same names, but that a specific identification is *permitted*.

Obligation 5 (Split indeterminacy budget). An indeterminacy budget I , separated into absorbable renormalization, blurring, and unabsorbed residual load.

Obligation 6 (Compatible evaluation in the container). A numerical evaluation ν_C in the container, *together with* the compatibility identities the claim needs — in particular $\nu_C \circ i_q = \nu_q$ on the stated source domain. These compatibility clauses are mandatory: the mere existence of a numerical map on C never discharges this obligation, and a stipulated invariance does not count as a proof of a compatibility.

Obligation 7 (Order compatibility). The implication itself: from the permitted linkage it *follows* that $\nu_q(\text{Obj}_q) \in H_\Theta$.

The load of the certificate sits almost entirely in the last two, and the distinction between them is the single most useful thing the instrument supplies. Obligation 6 asks whether the two sides are put on one scale by a stated compatibility — not merely housed in one container. Obligation 7 asks whether the order relation between the two evaluations is derived. The two can fail separately and for different reasons, and holding them apart is what turns a diffuse suspicion into two nameable missing statements. Without obligations 6 and 7, cohabitation in a common container is qualitative proximity, not the numerical statement that the argument needs.

Provenance of the sixth obligation’s wording. Version 0.1 of this paper stated obligation 6 disjunctively: a numerical evaluation in the container *or* a provable compatibility. The conjunctive form above is taken from the methods paper of the series [13], where the corresponding contract component was hardened after this paper first appeared. The change is not cosmetic — it is the reason two verdicts of version 0.1 are recoded here (“Changes in version 0.2”) — and it resolves a tension this paper carried from the start: the working lemma below already demanded a “label-compatible, scale-preserving” evaluation contract, which is more than the disjunctive form of obligation 6 required.

When an obligation counts as discharged. Because the obligations differ in kind, the evidence that discharges them differs too, and saying so in advance is what keeps the coding symmetric rather than convenient. Obligations 1–5 may be discharged by an explicit stipulation: a definition, a declared property, a fixed convention. Obligation 6 requires more than a shared symbol and more than a shared container: an inspectable assignment of both sides to one evaluation *and* the compatibility identities that tie that evaluation to the readouts it is supposed to represent. That both sides are real numbers is not one of them, and neither is a stipulated invariance. Obligation 7 is recorded as discharged only where the text contains a separately identifiable derivation, a named lemma, or an explicitly delegated result that yields the order relation from the permitted identification.

The negative form of that last rule is the one that constrains this paper most, and it is deliberately weak: failure to locate such an item, under the source range and the search terms reported with the finding, is a *documentary non-finding*. It is not a demonstration that no derivation exists, and it is not a judgment that the declared properties would be insufficient if the derivation were written out. That second question is mathematical, and Section 2.10 states that the instrument does not decide it. Table 2 provides a synoptic taxonomy of the seven formal obligations and the triggered extension, including their mathematical objects, operational requirements, corresponding ledger rows, and primary failure modes.

2.3 The ledger

The obligations are operationalized as nine rows (Table 3). Each row is a question to be answered against the wording of a primary text; each carries a recorded expectation, fixed

Obligation	Mathematical Object	Operational Requirement	Row	Primary Failure Risk
1 (q -readout)	(Obj_q, ν_q)	Explicit log-volumetric readout	R0, R2	Scale loss / implicit typing
2 (Θ -target)	$(\text{Obj}_\Theta, H_\Theta)$	Closed target half-line $H_\Theta \subset \mathbb{R}$	R0, R3	Moving target / post-hoc shift
3 (Container)	(C, i_q, i_Θ)	Shared container with typed embeddings	R1	Missing cohabitation
4 (Linkage)	$L \subset \text{Obj}_q \times \text{Obj}_\Theta$	Permitted identification (protocol)	R4	Unpermitted label confusion
5 (Budget)	$I = (I_{\text{abs}}, I_{\text{res}})$	Split into absorbed and unabsorbed load	R5	Hidden second mode
6 (Evaluation)	$(\nu_C, \nu_C \circ i_q = \nu_q)$	Container metric with mandatory compatibility	R6a	Unscaled container / missing identity
7 (Order)	$\nu_q(\text{Obj}_q) \in H_\Theta$	Derived inequality implication	R6b	Asserted membership / non-derivation
Ext. E8	Mode branch M_2	Triggered secondary evaluation channel	E8	Undisclosed branch selection

Table 2: Taxonomy of the seven bridge certificate obligations, formal signatures, and audit criteria.

before any text was examined, and a recorded risk — the failure mode that the row exists to catch. Recording the expectation in advance is not decoration: it is what allows a later reader to see which findings confirmed a prior belief and which contradicted one. The dating is part of the claim and is therefore stated: the obligations, the rows and their expectations were written down in May 2026; the four applications reported below were carried out in August 2026.

Three remarks on how to read this table. First, “open” in the expectation column means that no expectation was fixed, not that the answer was expected to be negative. Second, the rows are not of equal weight. As written, R6 names only the order question, but it in fact carries obligations 6 and 7 at once — a conflation which the split of Section 2.4 undoes, and which is worth naming, because while a row exists for the order question and none for the compatibility question, the second is assessed in running prose rather than at the ledger. R6 is the row on which the instrument was designed to be informative; R0, R1 and R3 are hygiene rows whose main function is to make a negative finding in R6 interpretable rather than generic.

Third, the last row is not a core row, and reading it as one distorts the denominator. R8 asks after a second evaluation mode, which is a feature of the domain model rather than of the contract: it maps to no obligation; it is active only where the claim’s domain actually offers several modes, branches or normalizations; and where that trigger is absent it is out of scope rather than negative. The successor instrument therefore carries it outside the obligation crosswalk, as the triggered extension E8, and reports a finding at that label and never at a contract component [13]. Nothing about the coding recorded for it below changes; what changes is its status, and that the eight core rows R0–R7 are the rows a completeness statement is about.

Version label and row namespace. The ledger stated here is the certificate ledger fixed in May 2026. Its rows are unprefixes, which was harmless while it was the only ledger of the series and is not harmless now: the successor instrument [13] scopes its rows by a version prefix (v1-T for the transport ledger, v1-F for the full one), and identical numeric suffixes across instruments do *not* imply identical semantics — this paper’s R8 is the second-mode row, whereas the row carrying that number in the transport instrument asks a different question.

Row	Test	Prior expectation	Risk the row catches
R0	Are the two lines separately typed?	yes	apparent equality of distinct things
R1	Is a common container present?	yes	a container does not supply a scale
R2	Is the numerical readout preserved on the q -side?	open	the readout loses its object identity
R3	Is the target half-line fixed on the Θ -side?	yes	the half-line is moved after the fact
R4	Is the j^2 -load addressed?	open	unrecorded $O(\ell^2)$ residual load
R5	Is the indeterminacy budget separated?	partly	blurring hides a second mode
R6	Is order preservation proved?	open	qualitative linkage is not enough
R7	Is approximation weaker than membership?	probably not, for a closed half-line	an apparent escape route
R8	Is a second mode visible or excluded?	open	the core of the question

Table 3: BC-Ledger v0: Nine audit rows with operational questions, prior expectations, and targeted failure risks.

Every unprefixed row label in what follows is a label of the ledger of this paper, and the correspondence with the successor rows is a crosswalk, not an identity. Where a version label is needed, this version assigns the ledger the name **BC-Ledger v0**; the label is introduced here and is not taken from the series register, which does not yet carry one for it.

2.4 Refinements produced by use

Three of the rows turned out to conflate distinct questions, and the instrument was sharpened accordingly. The first two refinements are *results of the applications* and were not part of the certificate as fixed; the third follows from the hardened reading of obligation 6 stated above. None of the three is a mere recommendation any longer: in the successor instrument all three are normative [13]. The case reports in Sections 3–6 stay with the original rows wherever the split makes no difference to the finding, and as a rule report at the split where it does — Case B at R4a and R4b, and Case A at R6a and R6b. Case C’s findings likewise separate at the two halves; its report retains the merged row of version 0.1, with the separation carried in the running text of Section 5.

R2a / R2b. Row R2 mixes two transports. *R2a* asks whether the readout survives transport *within* one container — along the link that the construction itself provides. *R2b* asks whether it survives the passage *between* containers. Only R2b is the bridge question proper. The distinction was forced by the comparative discrimination case of Section 5, where the two transports are visibly different in kind, and confirmed by Section 3, where R2b collapses into R6 because the identification of the containers *is* the bridge step.

R4a / R4b. Row R4 mixes the place where a load is quantified with the place where it is booked against the bridge. *R4a* asks whether the load is quantified in the evaluation of the Θ -side at all. *R4b* asks whether it is booked at the bridge itself. The distinction was forced by Section 4: the answer to R4a is yes and the answer to R4b is no, and a single word for both hides two different findings.

R6a / R6b. Row R6 mixes the two load-bearing obligations under a question that names only one of them. *R6a* asks after obligation 6: is the common evaluation accompanied by the compatibility identities that tie it to the two readouts? *R6b* asks after obligation 7: is the order relation derived from the permitted identification? This split was not produced by the applications but by the hardened reading of obligation 6 (Section 2.2). Once a common evaluation no longer discharges the sixth obligation by itself, a single row cannot carry both verdicts — Section 5 returns different answers at the two halves, and Section 3 returns the same answer for two different reasons; in either configuration a single merged row would have hidden the structure.

A further refinement is recorded as an open item rather than adopted: obligation 3 (container typing) currently enters the ledger only through R1, which asks whether a container is *present*. Both Sections 3 and 5 suggest that the interesting question is whether the typing is *sustained through the conclusion*, which is a different question and deserves its own row.

2.5 Verdicts and confidence

A row receives one of three verdicts, and the vocabulary is fixed because loose verdict language is how form-level findings turn into truth-level claims (see Table 4).

Verdict	Meaning
DELIVERS	The obligation is discharged by the wording examined.
PARTLY	Part of the obligation is discharged and a named part is not. The split must be stated, not averaged.
DOES NOT DELIVER	The obligation is not discharged by the wording examined. This is <i>not</i> a claim that the statement is false, nor that it cannot be discharged elsewhere.

Table 4: Verdict vocabulary for certificate evaluations.

Each verdict carries a confidence, and the confidence is about the *textual* finding, not about the mathematics: high confidence means that the quotations are unambiguous and that no verdict rests on a doubtful reading. A verdict may be high-confidence and narrow in reach at the same time; where that occurs, the reach restriction is stated next to the verdict rather than inside it.

These labels are author-assessed judgments about the textual evidence — the clarity of the quotation, the specificity of the anchor, the coverage of the search, and the reconstruction burden carried. They are not calibrated probabilities, and no independent second coding was carried out, so no inter-rater agreement is reported. Read them as a statement of how firmly the wording supports the entry, not as a measured reliability.

2.6 Gap typology

Gaps are typed, not graded. Grading (“serious”, “minor”) smuggles in an assessment the instrument is not entitled to make; typing states what kind of thing is missing, which is both more useful and more defensible (Table 5).

Types combine. A step whose repair would require both a missing argument and a resolution of a definitional ambiguity is recorded as G+D, and both parts must be named.

2.7 Import-ledger discipline

Every appeal to standard literature is registered as an *import*, with the exact locus in the published source. The rule attached to it is deliberately asymmetric:

Type	Description
G	A missing or non-reconstructible proof step.
Z	A citation that does not carry the statement it is invoked for — an <i>import</i> failure.
D	Definition drift: a term is used differently from the way it was defined.
N	Not an error, but a proof-relevant imprecision at the level of convention or notation.

Table 5: Typology of identified gaps and textual deficits.

Imports from peer-reviewed literature are treated as trustworthy. What is examined is their *fit* — whether the cited statement is the statement the argument needs — not their proof.

This is the operational content of the Z type. It keeps the audit finite: without it, every audit degenerates into an audit of the whole of arithmetic geometry. It also keeps the audit honest in the other direction, because an import whose fit was not checked must be recorded as unchecked rather than passed over.

2.8 Rules of evidence

Four rules govern what may be written into a ledger row.

No authority argument. Neither the examined author’s assertion that a step suffices nor a critic’s assertion that it does not is evidence. This applies in both directions and is not a courtesy: an author’s self-assessment is an object of examination, and a critic’s verdict is, for the purposes of the ledger, only the definition of what a negative control would look like.

Verbatim quotation with anchor. Every finding is carried by a quoted sentence with a line anchor into a named extract. A finding that cannot be quoted is not a finding.

“Searched, not found”. A claim that something is absent from a text is reported together with the search terms used and the range searched. This is the rule that makes negative findings falsifiable: a reader who suspects the search was too narrow can see exactly how narrow it was and repeat it. Every case report that rests on an absence claim carries such a block: Cases A, B and D below do. Case C makes no absence claim — its findings rest on quoted passages that are present — and therefore carries none.

Reconstruction discipline. Text extracted from PDF loses relation symbols, Greek letters, exponents and indices. Where a load-bearing sentence is affected, three things are required: the reconstruction must be stated explicitly, its derivation must be given from other passages of the same document, and a cross-check must be exhibited at an independent place. Where the naive reading of a damaged line would invert the meaning, that line must be flagged separately. This rule is part of the instrument, not an apology for it — and it matters here, because in one of the four applications a verdict does depend on such a reconstruction (Section 3.2).

2.9 The working lemma

The whole instrument is condensed in one statement, which is a task description and not a theorem.

Working lemma (bridge contract). For the forensic reading of a step of the shape described in Section 2, it does not suffice to exhibit a qualitative relation between the two objects inside a common container. What is required is an explicit, label-compatible, scale-preserving and order-compatible numerical evaluation contract Br_ξ . Any candidate for a second mode T_2 must either be visible inside that contract or be booked as an inadmissible hidden auxiliary scale.

Throughout, T_2 denotes *the second evaluation mode* — a second scale, shift or intertwining alongside the one the argument advertises — and nothing else. Row R8 is the ledger row that asks after it.

The status of this statement is: working lemma, not proved. It is to be treated as a precise task for primary-text examination and for positive and negative controls — which is what the remainder of this paper does with it.

2.10 What the instrument cannot decide

Stated positively, so that it is not mistaken for modesty.

1. **It cannot certify soundness.** All nine rows may deliver and the argument may still fail for a reason no row asks about.
2. **It cannot decide mathematical questions of substance.** Whether a pair of declared properties would suffice as a contract lemma if it were proved, or whether an identical global normalization of two isomorphic copies of one number field counts as a provable compatibility — these are mathematical questions. The ledger records only that the text supplies a property pair, or a convention, where it would need a lemma.
3. **It cannot see outside the documents examined.** A row returned as not delivered is a statement about a wording. The obligation may be discharged in material not examined; where that possibility is concrete, the case reports name it.
4. **It cannot distinguish an unmet obligation from an unmeetable one.** Nothing in the instrument licenses the step from “not derived here” to “not derivable”.

2.11 Applying the certificate to a new argument

The instrument is not tied to the four cases below. The following recipe is a prospective protocol distilled from those applications. It was not completely executed in every reported case: in particular, Case A leaves hygiene rows open, and the hygiene-before-R6 rule was formulated after the applications. The recipe is offered for a new transfer argument; the historical partial applications do not retrospectively satisfy all its steps.

- Application recipe.**
1. **Locate the step.** Identify the sentence at which a qualitative relation becomes a numerical assertion. This is usually not where the document’s own summary says it is; verify the structure against the text and record the correction, because the next reader should not have to redo it.
 2. **Name the two sides and the container.** Fill obligations 1–3. Record the typing of the embeddings, and record whether the typing is dropped anywhere before the conclusion.
 3. **Fill the hygiene rows R0, R1, R3.** If one of them fails, a later finding at R6 may be an artifact of the failure rather than an independent result.

4. **Separate obligation 6 from obligation 7.** Ask first whether the container carries a numerical evaluation *and* the compatibility identities that tie it to the two readouts — in particular $\nu_C \circ i_q = \nu_q$ on the stated source domain. A common evaluation alone never discharges obligation 6, and a stipulated invariance is not a proof of a compatibility. Only then ask whether the order relation is derived from the permitted identification. Conflating the two questions is the most common way to produce a finding that is both unfair and uninformative; collapsing the first into the bare existence of a common scale is the most common way to produce one that is too generous.
5. **Search for the contract lemma explicitly**, with recorded search terms, over the whole document and over the proof range separately. Record the result either way.
6. **Type any gap found** (Section 2.6) and enter every appeal to external literature into the import ledger (Section 2.7).
7. **Write the “searched, not found” block** before writing the verdict, not after. Its function is to constrain the verdict, which it cannot do if it is composed to fit one.
8. **State the reach separately from the verdict.** “Does not deliver, high confidence, restricted to document X ” is a complete finding; “does not deliver” alone is not.

3 Case A: the corollary and its proof

The first application is to the passage the instrument was built for: the proof of Corollary 3.12 in the third volume of inter-universal Teichmüller theory [1]. Rows R2, R4, R6 and R8 were filled — R6 at both halves of the split introduced in Section 2.4, and R8 as the triggered extension row it is rather than as a core row; R0, R1, R3, R5 and R7 were not part of this application and remain open (Section 7).

Relation to Part 1 of the series. This passage has already been located in print. An earlier, repair-oriented study of the same program [12] places the gap at the transition between the same substeps and reports the same wording — that the required inequality is said to “follow formally”. The present application neither repeats nor strengthens that diagnosis, and it claims no priority over it. What it does is narrower: it recodes the documentary question as one obligation of a stated certificate, so that the same site can be compared with three further texts under one unchanged instrument. That two routes of a different kind arrive at the same address is recorded here as a finding about the address, not as a second detection.

This is therefore a *partial* ledger application, and it is reported as one. The recipe of Section 2.11, which asks for the hygiene rows R0, R1 and R3 to be filled before R6 is weighed, was written afterwards, out of the experience of these applications — as were the row splits of Section 2.4, and for the same reason the case reports are not retrofitted to it. The consequence is stated rather than absorbed: what follows is a finding about four rows. The confidence recorded with each of them is confidence in that row, and no full-certificate verdict on this passage is claimed here.

3.1 Locating the passage

The substeps in which the transition occurs stand *inside the proof* of the corollary, not in the remark that recapitulates it. The structure, recorded here so that it need not be re-derived (Table 6):

Four properties declared in Remark 3.11.1 recur throughout and are abbreviated as in the source: the input prime-strip link (IPL), the strip-holomorphic expressibility condition (SHE), algorithmic parallel transport (APT), and hidden internal structures (HIS).

Block	Extract lines
Proposition 3.9 (log-volume for packets and processions)	6365 ff.
Theorem 3.11 (multiradial algorithms), proof	8666 ff., proof 9032
Remark 3.11.1 with the declared properties	9036–9300
Corollary 3.12, statement	9851–9884
Corollary 3.12, proof	9886 ff.
Step (x) (indeterminacy balance)	10237–10283
Substeps (xi-a)–(xi-h)	10285–10496
Remark 3.12.2 including the toy model	10573 ff., toy model 10733–10813

Table 6: Structural blocks and line anchors of the IUT-III Corollary 3.12 proof extract.

3.2 Reconstruction disclosure

The extract of this document loses relation symbols, exponents, indices and Greek letters. Unlike in the other three applications, one verdict here *does* depend on a reconstruction, so the derivation is given in full, per Section 2.8.

1. The target set is defined at l. 10385: the extract reads $R - |\log()| \text{ d=ef } \{ R | -|\log()| \} R$, to be read as $\mathbb{R}_{\leq -|\log(\Theta)|} := \{\lambda \in \mathbb{R} \mid \lambda \leq -|\log(\Theta)|\} \subset \mathbb{R}$.
2. The corollary’s own statement resolves the direction of the relation at l. 9882/9884: “... i.e., $C_\Theta \geq -1$ for any real number $C_\Theta \in \mathbb{R}$ such that $-|\log(\Theta)| \leq C_\Theta \cdot |\log(q)|$ ”, hence $-|\log(\Theta)| \geq -|\log(q)|$.
3. Together these make l. 10446/10448 unambiguous: $-|\log(q)| \in \mathbb{R}_{\leq -|\log(\Theta)|}$, “hence also the inequality” $-|\log(q)| \leq -|\log(\Theta)| \in \mathbb{R}$.

The disputed membership therefore stands as an *explicit statement* at l. 10446 — with the relation symbols restored as set out above — and is not an interpretive inference of ours.

Verification against the page images. Because a verdict of this section depends on the restoration, the two affected lines were checked against the published pages themselves rather than against the extract alone. Both restorations are confirmed. On p. 184 the page carries the membership $-|\log(q)| \in \mathbb{R}_{\leq -|\log(\Theta)|}$ and, displayed beneath it, $-|\log(q)| \leq -|\log(\Theta)| \in \mathbb{R}$, followed by “then follows formally”; on p. 174 the corollary’s statement carries $-|\log(\Theta)| \geq -|\log(q)|$ together with the sentence quoted above. The symbols the extract dropped are therefore present in the source, and in the form the reconstruction assumed. This also corrected one detail of our own transcription: the source writes C_Θ throughout, not a bare C , and the quotations here now follow it.

One further line is flagged because the naive reading inverts its sense. At l. 10750 the extract reads **in the sense that** $R -h = -2h R$; the intended reading is $\mathbb{R} \ni -h \neq -2h \in \mathbb{R}$, the “ $\neq$ ” having been lost. The surrounding sentence secures this: the two assignments are “mutually incompatible and cannot be considered simultaneously” (IUT-III, extract l. 10748–10749). A reader who restores “=” here would conclude the opposite of what the passage says — and the page image settles it: p. 190 carries “[in the sense that $\mathbb{R} \ni -h \neq -2h \in \mathbb{R}$]”.

3.3 R2 — is the q -side readout preserved?

DELIVERS, with high confidence — and the reason is structurally interesting: at the decisive step no transport takes place at all.

The q -side is declared indeterminacy-free in the statement of the corollary itself:

“for the procession-normalized mono-analytic log-volume of the image of a q -pilot object ... in the multiradial representation of Theorem 3.11, (i), *which we do not regard as subject to the indeterminacies* (Ind1), (Ind2), (Ind3) described in Theorem 3.11, (i), (ii).” (IUT-III, extract l. 9869–9873)

The immediately preceding paragraph attributes exactly the opposite to the Θ -side (IUT-III, extract l. 9862–9864). The asymmetry is thus *declared*, not accidental. The q -pilot is moreover the fixed reference point of the construction — property (IPL) calls it “the ‘coric’/‘fixed’ q -pilot $\mathcal{F}^{\text{lx}}$ -prime-strip” (IUT-III, extract l. 9117–9122) — and property (APT) states that there is no set-theoretic transport to speak of:

“This parallel transport mechanism *does not consist of a simple instance of transport of some set-theoretic region ... via some set-theoretic function*. Rather, it consists of a construction algorithm that is simultaneously valid/executable/well-defined with respect to the arithmetic holomorphic structures in the domain and codomain of the Θ -link” (IUT-III, extract l. 9212–9219)

The readout is concrete: $-|\log(q)|$ is “the log-volume of the region that arises from the representation of the q -pilot object” (IUT-III, extract l. 10387–10390), on a scale anchored by the statement that “global arithmetic degrees of objects of global realified Frobenioids may be interpreted as log-volumes” (IUT-III, extract l. 10391–10393).

Obligation 1 is therefore discharged, and obligation 4 is discharged on the q -side — but *stipulatively*: the identity of the pilot is protected by the declarations “coric”/“fixed” and “do not regard as subject to” rather than produced by a proved relation.

3.4 R4 — is the j^2 -load addressed?

PARTLY, with high confidence. The place of the booking is named; the load itself is not quantified there, and its quantification is explicitly delegated to the fourth volume.

The load exists and is named: the relevant monoids are generated by the theta values $\{q^{j^2}\}_{j=1,\dots,\ell^*}$ (IUT-III, extract l. 654–656). The booking site is the procession normalization, and what it books is *labels*:

“the procession-normalized mono-analytic log-volumes [i.e., where the average is taken over $j \in \mathbb{F}_\ell^*$] ... are *invariant with respect to the indeterminacies* (Ind1), (Ind2), and have the effect of *converting the indeterminacy* (Ind3) *into an inequality [from above]*.” (IUT-III, extract l. 10261–10268)

with the averaging attributed to “the $\mathbb{F}_\ell^*$ -symmetry acting on the labels of the theta values” (IUT-III, extract l. 10273–10276). That is a booking of symmetry, not of magnitude.

For the corollary the load is then explicitly set aside, and the delegation is declared openly. Property (HIS) names this very application:

“if one takes the point of view — *as will be done in Corollary 3.12 below!* — that one is only interested in considering the *qualitative logical aspects/consequences* ... then ... one may ... *forget that this construction algorithm ... has anything to do with theta functions ... or theta values* [i.e., the q^{j^2}] $_{j=1,\dots,\ell^*}$!” (IUT-III, extract l. 9225–9244)

and, immediately afterwards:

“unlike Corollary 3.12 below, which only concerns qualitative logical aspects/consequences ..., the *explicit computation to be performed in* [IUT-IV], §1, *of the log-volumes that occur in the statement of Corollary 3.12 makes essential use of the way in which theta values occur* in the construction algorithm of Theorem 3.11.” (IUT-III, extract l. 9263–9267)

A search of the whole proof range for the theta values, the procession normalization and the averaging returns hits *only* in Step (x); in the substeps themselves the load does not reappear.

One point of fairness belongs to the verdict rather than beside it. The failure mode this row exists to catch — a load that vanishes in one coordinate and returns without being booked — *does not occur here*. What occurs is a declared division of labour between a qualitative volume and a quantitative one. For the certificate the finding remains that within the document examined the load is not booked at the disputed place; the distinction between “booked elsewhere, by declaration” and “concealed” is recorded in the source’s favor.

3.5 R6 — is order preservation proved?

DOES NOT DELIVER at both halves of the row, with high confidence. This is the core finding of the application, and it must be stated in two parts, because the two obligations fail for different reasons and the row as originally worded names only the second of them.

R6a — obligation 6 is not discharged: a common evaluation, no compatibility identity. In substep (xi-d) the two sides are explicitly brought into one evaluation:

“it is only by forming [a suitable positive tensor power of] the determinant of the localizations of arithmetic vector bundles . . . and then applying the [suitably normalized . . .] log-volume to various regions . . . that we are able to obtain *completely comparable objects*, namely, $\mathbb{R}_{\leq -|\log(\Theta)|} := \{\lambda \in \mathbb{R} \mid \lambda \leq -|\log(\Theta)|\} \subset \mathbb{R}; -|\log(q)| \in \mathbb{R}$ ” (IUT-III, extract l. 10376–10385)

Together with the log-volume interpretation of arithmetic degrees quoted above, a numerical evaluation ν_C exists in the container, and it is stated explicitly rather than left to be inferred. That much is recorded in the source’s favor. It is not, however, what obligation 6 asks for. The obligation asks for the compatibility identity $\nu_C \circ i_q = \nu_q$ across the identification of the two containers, and that identity is not in the text: the “searched, not found” block below records the search, and records that the one compatibility lemma which does exist — the log-link compatibility of the log-volumes — secures compatibility *within* a vertical column rather than across the identification. Since a common evaluation never discharges obligation 6 by itself (Section 2.2), R6a does not deliver.

Version 0.1 of this paper coded this half the other way, on the disjunctive wording of obligation 6 then in force, and stated that “the gap sits at obligation 7, not at obligation 6”. The quotations are unchanged and no source was re-examined; the coding follows the hardened wording. See “Changes in version 0.2”.

R6b — obligation 7 is set as a condition, not derived. Substep (xi-e) renders (SHE) operationally as a side condition on the admissible operations:

“‘expressibility relative to the arithmetic holomorphic structure in the 1-column’ [cf. (SHE)] amounts precisely to ‘*expressibility via operations that are valid/executable/well-defined even when subject to the condition that the pilot-object log-volume associated to the input data prime-strip . . . be equal to the fixed value $-|\log(q)| \in \mathbb{R}$* ’” (IUT-III, extract l. 10412–10420)

and substep (xi-f) draws the membership from it:

“the construction of the subset $\mathbb{R}_{\leq -|\log(\Theta)|} \subset \mathbb{R}$ of possible pilot-object log-volumes of output data *is subject to the condition* that this construction of output data possibilities *constitutes, in particular, a construction* [perhaps only up to some sort of ‘approximation’, as a result of various indeterminacies] *of the pilot-object log-volume of the input data*

prime-strip, namely, $-|\log(q)| \in \mathbb{R}$. The inclusion $-|\log(q)| \in \mathbb{R}_{\leq -|\log(\Theta)|}$, hence also the inequality $-|\log(q)| \leq -|\log(\Theta)| \in \mathbb{R} \dots$ then follows formally.” (IUT-III, extract l. 10437–10451)

Substep (xi-g) names the result as “two *tautologically equivalent* ways to compute the log-volume of the q -pilot object” (IUT-III, extract l. 10453–10458), and the author’s own recapitulation repeats the pattern: the q -intertwining “may be thought of as a special case of the Θ -intertwining ... Corollary 3.12 then follows, essentially formally” (IUT-III, extract l. 10723–10731).

The toy model writes the carrying assumption out as an assumption. Without labels the two assignments contradict one another (IUT-III, extract l. 10747–10750); the decisive step is then introduced as a supposition:

“Now suppose that one verifies that one may construct the ‘assignment with indeterminacies’ ... as a mathematical structure that is intrinsically associated to the underlying structure of the assignment Ξ_q ... even if one forgets the labels ‘ q ’, ‘ Θ ’ ... i.e., even if one identifies $q\mathbb{R}$, $\Theta\mathbb{R}$, in the usual way, with $\mathbb{R}$ [cf. the properties (IPL), (SHE)]” (IUT-III, extract l. 10778–10784)

from which “one then concludes formally” the desired bound (IUT-III, extract l. 10806–10813).

The finding, in certificate language. In the real case the supposition is met by the conjunction $(\text{IPL}) \wedge (\text{SHE})$ — that is, by *properties attributed to the output of an algorithm*, not by a compatibility lemma $\nu_C \circ i_q = \nu_q$. At the same place the container typing required by obligation 3 is deliberately released: the identification of $q\mathbb{R}$ with $\Theta\mathbb{R}$ is a premise of the inference, not its result. The absent item is thus one and the same at both halves of the row, and it is absent twice over: the identity is what obligation 6 requires in order to put the two readouts on one scale by proof rather than by cohabitation, and it is what obligation 7 would need in order to carry the order relation across that scale.

Fairness: “tautologically” is meant affirmatively. The source uses this word approvingly, not as a concession: it speaks of concluding “nontrivial, albeit essentially tautological, consequences ... such as the inequality of Corollary 3.12” (IUT-III, extract l. 8122–8125). For its author the tautologous character is the achievement, not the defect. Reading the word as a self-refutation would be an authority argument in reverse, and is not done here. The ledger asks only whether the compatibility identity is stated and whether the order relation is *derived*; by the wording, neither is.

3.6 R8 — is a second mode visible or excluded?

PARTLY, with high confidence. Named and qualitatively priced, but not carried through the bridge as a second scale; not excluded either.

The double structure is stated explicitly: a prime-strip “*simultaneously equipped with two intertwining*s, namely, the Θ -intertwining, *up to indeterminacies*, and the q -intertwining” (IUT-III, extract l. 8114–8125). The price of the construction is declared as well:

“this ‘solution’ may be implemented *only at the cost of admitting the ‘indeterminacy’ constituted by the upper semi-compatibility of (Ind3)*.” (IUT-III, extract l. 10019–10031)

and the content of that notion is given by the text itself: “upper semi-compatibility, i.e., in a word, *one-sided inclusions, as opposed to precise equalities*” (IUT-III, extract l. 8489–8490).

What does not happen is that the second mode is carried through the disputed step. Before the bridge it is resolved into a direction — (Ind3) is converted “into an inequality [from above]” — and at the bridge the internal structure is, by (HIS), expressly forgotten. Visible before the step, suspended within it.

A near-miss is recorded rather than suppressed: a remark whose title (“The log-shifted nature of the localization homomorphism”) suggests a visibly carried second scale concerns only the localization homomorphism of an earlier proposition (IUT-III, extract l. 5551–5553) and does not support the row.

3.7 Result of Case A

The resulting findings across the audited rows are summarized in Table 7.

Row	Finding	Confidence
R2	DELIVERS — readout definitionally fixed and indeterminacy-free; no transport step	high
R4	PARTLY — site named, load not quantified, quantification delegated	high
R6a	DOES NOT DELIVER — common evaluation present and explicit; compatibility identity $\nu_C \circ i_q = \nu_q$ absent	high
R6b	DOES NOT DELIVER — membership set via a property pair; follows “formally”	high
R8	PARTLY — named and priced, suspended at the bridge	high

Table 7: Case A: Summary of findings across audited BC-Ledger rows.

The address of the two unmet obligations is a single substep. There, the membership “then follows formally” from the condition that the construction of the output possibilities *also* constitutes a construction of the input readout (IUT-III, extract l. 10437–10451). What is absent is a compatibility identity $\nu_C \circ i_q = \nu_q$ across the container identification, which obligation 6 requires in its own right, and, on top of it, a contract lemma that *derives* the order relation from the permitted linkage, which obligation 7 requires. In their place stands the property pair, whose discharge the toy model writes out as a supposition. What the text does deliver at this substep is the common evaluation itself, explicitly — and that is not nothing; but a common evaluation without the identity is exactly the cohabitation the certificate was built to keep apart from a numerical statement.

Searched, not found. Over the whole extract and, separately, over the proof range: a compatibility lemma of the form $\nu_C \circ i_q = \nu_q$ across the container identification (search terms: `compatib`, `log-volume`, `equal`, `identif`); a numerical value for the approximation term (`approximation`, `epsilon`, `bound`, `estimate`, `Ind3`); an explicit booking of the $O(\ell^2)$ load at the bridge (`theta value`, `procession-normalized`, `average`); a visibly carried second evaluation scale at the bridge (`log-shift`, `vertical shift`, `two intertwining`s). One important qualification, recorded so that the negative finding is not read as carelessness: a genuine compatibility lemma *does* exist — the log-link compatibility of the log-volumes (IUT-III, extract l. 10268–10272) — but it lies on the wrong axis. It secures compatibility *within* a vertical column, not across the identification of the two evaluation containers — which is what obligation 6 asks for, and what obligation 7 would then have to build on.

Reach. High confidence attaches to the textual findings, row by row, in the author-assessed sense fixed in Section 2. The reach is restricted twice over: only the third volume was examined here, and only four of the nine rows were filled, so what is reported is a finding at four rows and not a verdict of the whole certificate. The quantitative continuation named by the source itself is the subject of Case B.

4 Case B: the declared quantification site

Case A left one concrete possibility open: the load that is not booked at the bridge might be booked at the site to which the text explicitly delegates it. The second application opens that site [2] and asks two strictly separated questions.

F1 Is the j^2 /procession load actually quantified there?

F2 Does that section *derive* the disputed membership, or does it *presuppose* the corollary?

Keeping them apart is essential, because a positive answer to F1 is easily mistaken for progress on F2, and it is not.

4.1 F1: quantified

QUANTIFIED, with high confidence. The chain is written out in the source and is here only exhibited, not recomputed.

The theta value enters as the input quantity of the computation: “ λ ” is set to be “ $\text{ord}(-)$ ” of the element $q_{v_j}^{j^2}$ at the bad places (IUT-IV, extract l. 1594–1597). The local term carries j^2 visibly, as $j^2/(2\ell(j+1))$ (IUT-IV, extract l. 1656–1657); after weighting, $j^2/(2\ell)$ remains (IUT-IV, extract l. 1669–1670). The procession averaging then evaluates the sum of squares explicitly:

$$\begin{aligned} &\text{“it remains to compute the average over } j \in \mathbb{F}_\ell^*. \text{ By averaging over } j \in \{1, \dots, \ell^* = \\ &(\ell-1)/2\} \text{ and applying (E1), (E2), we obtain the ‘procession-normalized upper bound’ } \dots \\ &= \frac{\ell+5}{4} \log(d_K) - \frac{\ell+1}{24} \log(q) + \dots \text{”} \end{aligned} \quad \text{(IUT-IV, extract l. 1675–1684)}$$

the identity applied being $\frac{1}{n} \sum_{m=1}^n m^2 = \frac{1}{6}(2n+1)(n+1)$ (IUT-IV, extract l. 1379–1380). After the local estimate the coefficient survives into the final bound, where it appears as $-\frac{1}{6}(1 - \frac{12}{\ell^2}) \log(q)$ (IUT-IV, extract l. 1775–1776), and in the statement of the theorem itself (IUT-IV, extract l. 1352–1353). The residual ℓ -dependence is named and discussed as “an inevitable consequence of the fundamental role played ... by the ℓ -torsion points” (IUT-IV, extract l. 1909–1912).

The chain as displayed leaves no term unaccounted for: the load arises only at the bad places, the good non-distinguished places contribute zero, the archimedean step is computed separately, and the final step sums them.

4.2 F2: presupposes

PRESUPPOSES, with high confidence, and the strongest evidence is negative. Three complete searches over the whole document return: zero hits for the q -pilot; zero hits for each of the four declared properties on which the bridge rests in Case A; zero hits for the intertwining. All occurrences of “pilot” in this document are Θ -pilots. A document that, under the terms searched, never names the q -pilot and never mentions the properties carrying the bridge supplies no derivation of the bridge. The unmet obligation of Case A lies outside this section’s subject matter, as far as those terms reach.

The framing says the same thing positively. The section is introduced as producing “*more explicit versions ... of the log-volume estimates for Θ -pilot objects obtained in [IUT-III], Corollary 3.12*” (IUT-IV, extract l. 456–458), and its main theorem opens with “*Suppose that we are in the situation of [IUT-III], Corollary 3.12*” (IUT-IV, extract l. 1281).

So does the proof logic. The section supplies the upper side — “we thus obtain the following *upper bound*” (IUT-IV, extract l. 1773), “substituting back ... we obtain the following *upper bound for ‘C’*” (IUT-IV, extract l. 1821) — and imports the lower side:

“and hence, *by applying the inequality ‘ $C_\Theta \geq -1$ ’ of [IUT-III], Corollary 3.12, conclude that ...*” (IUT-IV, extract l. 1355–1356)

That imported inequality *is* the disputed membership in constant form: the corollary’s conclusion states $-|\log(\Theta)| \geq -|\log(q)|$, “i.e., $C_\Theta \geq -1$ for any real number C_Θ such that $-|\log(\Theta)| \leq C_\Theta \cdot |\log(q)|$ ” (IUT-III, extract l. 9878–9884). The architecture the text writes down for itself is therefore: this section bounds C from above, the corollary bounds it from below, and only both together say anything about $\log(q)$. An upper bound on C alone says nothing.

Finally, the section does not even handle the third indeterminacy itself. At all three places where it arises, the treatment is delegated backwards: it “is taken into account by the fact that we are considering upper bounds [cf. the discussion of Step (x) of the proof of ... Corollary 3.12]” (IUT-IV, extract l. 1609–1611).

4.3 Consequence for the ledger

Three consequences, and only the first is a change to a verdict.

1. **Row R4 splits.** The load is quantified in the evaluation of the Θ -side (R4a: yes, in the combined reading of the two volumes) and is not booked at the bridge (R4b: no in the third volume, out of scope in the fourth). Without the split, one word hides two findings.
2. **The delegation is verified.** The concern that motivated the row — a load that disappears in one coordinate and returns unbooked — does not apply to the two volumes read together: the theta value disappears in the third and *returns named and computed* in the fourth, at exactly the place the third had specified. This is recorded in the source’s favor.
3. **The R6 verdict is unchanged at both halves, and its reach is extended.** Case A had stated its verdict under the reservation that only one volume was examined. For the site named by that volume, the reservation is now discharged: there is neither a compatibility identity nor a contract lemma there, and no candidate for either. The site is not contradictory but thematically uninvolved.

One further observation belongs here, as an observation about architecture rather than a judgment: by taking $C_\Theta \geq -1$ as an input, the fourth volume makes visible how much hangs on the membership. Its entire explicit computation produces an upper bound which, without the lower bound supplied by the corollary, yields no statement about $\log(q)$ at all.

Searched, not found. A derivation of the membership anywhere in this volume (search terms: `q-pilot`, `pilot`, `holomorphic hull`, `inclusion`, `follows formally`, `tautolog`, `Step (xi)`): not found; the one hit for the substep lies in a later section and refers back to the third volume. A contract lemma or any discussion of the property pair: not found, zero hits. A numerical value for the toy model’s approximation term: not found — and the tempting identification of that term with the bound computed here is *our* association, not a finding of the text, which nowhere makes it.

Import ledger. The verdict of F2 turns on a single appeal to outside material, and the discipline of Section 2.7 requires that it be entered rather than passed over (Table 8).

Field	Entry
Importing passage	IUT-IV, § 1, in the proof of its main estimate (IUT-IV, extract l. 1355–1356)
Statement imported	the inequality “ $C_\Theta \geq -1$ ” of IUT-III, Corollary 3.12
Statement needed	the same inequality: this section produces an upper bound for C and needs a lower bound it does not derive
Fit	exact — imported statement and required statement coincide
What was checked	the fit, not the proof, per the asymmetric rule of Section 2.7

Table 8: Case B: Import ledger entry for the IUT-IV main estimate.

No type Z failure is recorded here: the import carries exactly the statement it is invoked for. That is precisely why the entry matters. The statement imported is the one whose derivation obligation 7 asks after, so the appeal is sound and the site still cannot discharge the obligation.

5 Case C: a comparative discrimination case in a different theory

The third application is not a second test of the same text. It applies the unchanged instrument to an independent alternative program [5, 6] which reaches the same inequality by different means. Its function is to answer a question that a single application cannot: does the certificate return the same finding everywhere — in which case it is a template, not an instrument — or does it discriminate?

The label matters and is fixed here in the terms of the methods paper [13]: this is a *comparative discrimination case*, not a control. Version 0.1 called it a comparison control, which claimed more than the case can carry — a control would have to be a transfer argument whose seventh obligation *is* discharged, and Section 7 already recorded that no such case was among those examined. The function described below is unchanged; only the name is corrected.

It discriminates. Three of the four rows deliver, the fourth delivers only in a restricted sense, and the unmet obligation sits at a different address and has a different shape. *Where* that unmet obligation sits was itself revised by two later corrections (Section 5.5); the comparison at the end of this section states the revised reading rather than the first coding, and the revision is reported as part of the case.

5.1 R2: delivered, by a proved relation

The transport along the construction’s own link preserves the readout, and the j^2 factor is carried visibly: the scaling relation between the valuations is stated as part of a theorem, $|-|_{K'_{y_w,j}} = |-|_{K'_{y_w,1}}^{j^2}$ for $j = 1, \dots, \ell^*$ (ATS-III, extract l. 1544–1560), and is used in the explicit computation of the theta values’ Bloch–Kato classes (ATS-III, extract l. 6458–6484). The evaluation scale is concrete: a log-volume with a stated normalization (ATS-III, extract l. 6562–6596). Obligation 1 is discharged; obligation 4 is discharged here by a *proved valuation relation*, not by a stipulation.

This is the contrast that forced the R2a/R2b split. A second transport also occurs, between two copies of the ambient data related by a global Frobenius, and there the identification is *set by a normalization choice*: “Choose the same global normalization ... With this choice ... one

has” — this in the *statement* of Theorem 6.10.1 of ATS-IV (ATS-IV, extract l. 3186–3189) — the proof of the resulting equality being, in that theorem’s proof, the single sentence that it “is a consequence of the choice of the same global normalization” (ATS-IV, extract l. 3262–3266). Same row, two different kinds of answer.

5.2 R4: delivered, by a global constraint

The load arises only at the semistable places, survives into the numerical evaluation, and its compensation is booked — the booking instrument being the product formula:

“the non-trivial scaling relationship ... at primes $w \in V^{\text{odd},ss}$ *forces that suitable (re)normalization of valuations must be introduced* at primes $w \in V_{L'} - V^{\text{odd},ss}$ so that the product formula ... hold[s] for each normalized arithmeticoid” (ATS-III, extract l. 1600–1609)

Honesty requires the limitation alongside: the proof of that statement is one sentence, so the *existence* of a consistent compensating renormalization is inferred from the constraint rather than constructed. The load is located, not quantified. The failure mode the row exists to catch does not occur.

5.3 R6: partly — proved as an inequality, conventional as a bridge

The inequality itself is a theorem with a proof — Theorem 9.11.1 of ATS-III, in its §9.11 (ATS-III, extract l. 6794–6806) — and the evaluation in the container is genuinely numerical, computed as a sum of local contributions (ATS-IV, extract l. 764–769). Obligation 6 is *partly* discharged: the evaluation is numerical, but the compatibility holds only between the sums and is place-wise disclaimed by the source itself — as the passages quoted in the next paragraph state in the source’s own words. Version 0.1 coded this half as discharged “via its first alternative”; under the hardened wording of obligation 6 the same quotations yield the partial code, and the reasoning was already in the text (“Changes in version 0.2”).

The bridge that carries the final conclusion is a different matter. Two evaluations, in two copies related by the global Frobenius, are joined by a middle equality whose proof, in the proof of Theorem 6.10.1, reads in full:

“The middle equality is a tautology using the fact that the number fields provided by y'_0 and y_0 are identically normalized.” (ATS-IV, extract l. 3272–3273)

while the same text expressly forbids the place-by-place comparison a reader might use to check it: “one cannot claim that $z_p \leq z'_p$ holds for all primes p without violating the middle equality” (ATS-IV, extract l. 816–819), and the difference by Frobenius at each prime “*precludes direct comparison of various local quantities*” (ATS-IV, extract l. 3208–3211). Order compatibility thus exists only between the sums, never place by place.

5.4 R8: delivered — and this is why the rest is legible

The second mode here is the global Frobenius shift, and it is not merely present but declared, used asymmetrically, interpreted, and flagged as a correction of the author’s own earlier version: the lower bound is computed in one copy and the upper bound in the other, “the upper bound is not independent of this choice” (ATS-IV, extract l. 733–743), and a bracketed note records that “[t]he previous release of this paper had omitted this Frobenius-shifting” (ATS-IV, extract l. 755).

This is the strongest positive result of the comparison, and it has an analytic payoff rather than a rhetorical one: *because* the second mode is disclosed rather than absorbed into a normalization, the gap at R6 has an address at all.

5.5 Two later corrections, pulling in opposite directions

Both must be reported, or the case is misrepresented.

Strengthening. A source examination of the two earlier volumes of the same series, published separately as a tool supplement of this series [15], settled the definitional question left open above: the normalized object is unique per point, so “the same global normalization” is *definitional*. On that reading the identification is not load-bearing, and the middle equality rests instead on the by-construction Frobenius stability of the collated adelic set — that is, on precisely the mechanism this application classified as delivered under R8 — together with a definitional pinning of the scale. What remains is not a missing contract lemma but a narrower question about a measure convention.

Weakening. A substance examination of the theorem carrying R6 — companion work of this series, published as Part 5 [14] — found that its core step, a containment declared “clear from the construction”, is not carried by the construction cited for it. That is a G+D gap: a missing argument (a supremum statement is used where a measure statement is needed) together with a structural ambiguity about where one algebraic structure becomes another. The repair duty is named; the finding is not an error claim.

The two together are best summarized in the words those two examinations used: the difference in *form* remains, and the advantage in *substance* shrinks.

The revised coding. Taken together, the two corrections do not make this case better or worse than the first coding recorded; they *move* the deficit. Rows R2, R4 and R8 are untouched by either correction and continue to deliver. What changes is the split inside the PARTLY of row R6. Before the corrections, the delivered part was the inequality as a proved theorem and the missing part was a bridge joined by a conventional identity. After them, the identity is definitional and therefore not the missing part, while the missing part sits inside the theorem itself, at its containment step. The verdict stays PARTLY because the rule is that a split verdict must state its split rather than average it — and it is the split that has changed, not the verdict.

That a later and deeper reading revised the study’s own first coding is reported here rather than smoothed away. An entry that a subsequent examination of the same sources can correct is an entry that can be checked; that is a property worth having. It is not a claim that the instrument has been validated.

5.6 What the comparison shows

Both arguments fail obligation 7 in the strict sense of the ledger, and they fail it differently — which is the whole point of a comparative discrimination case (Table 9).

The difference that matters is the *shape* of the unmet obligation, and the corrections above have changed where that shape sits in Case C without changing that it is one identifiable step. In Case C the deficit is localized: after the corrections it sits at a single containment step inside the theorem that carries the inequality, where before it was read as sitting at the conventional identity joining the two evaluations. In Case A it is areal: a conjunction of two declared properties, distributed over a remark and four substeps, discharged by an inference the text itself calls formal. The contrast is between a deficit at one identifiable step and a deficit distributed over several. It is not a statement about which is more serious — the typology of Section 2.6 types gaps and deliberately does not grade them.

Row	Case A	Case C
R2	delivers — no transport occurs (declared)	delivers — proved valuation relation; second transport conventional
R4	partly — site named, quantification delegated	delivers — booked via the product-formula constraint
R6	does not deliver at both halves — no compatibility identity, and membership “follows formally”	partly, revised — compatibility only between the sums, place-wise disclaimed; inequality carried by a theorem, deficit inside that theorem’s containment step
R8	partly — named, then suspended	delivers — disclosed and used asymmetrically

Table 9: Comparative discrimination matrix: BC-Ledger findings in Case A (IUT-III) versus Case C (ATS-IV).

6 Case D: downstream applications

The fourth application asks a different question of two papers that *use* the disputed inequality [7, 8]: do they presuppose the certificate, circumvent it, or introduce an alternative scale? A finding here cannot detect the gap — it can only show how the gap propagates.

A note on the label. These papers are filed in the underlying material under a name that does not appear in them; the curves they use are Frey–Hellegouarch curves, and the author is a single author. The label is an archival artifact and is not used here.

6.1 Where the bound enters

In the first paper the disputed inequality appears as Proposition 1.9, its formula (1.7), whose entire proof is one sentence: it “follows from the μ_6 -version of” the corollary of Case A (Zhou-I, extract l. 728–732). The formula number is used exactly once afterwards, in the proof of the next proposition (Zhou-I, extract l. 753–757), and Proposition 1.9 is never revisited. There is no derivation, no intermediate lemma, and no contract lemma of the kind obligation 7 requires.

Everything the paper contributes of its own lies downstream, on the Θ -side: a sharper hull bound (Zhou-I, extract l. 482–496), its own readout of the Θ -pilot (Zhou-I, extract l. 677–725), and its own constants in the target inequality (Zhou-I, extract l. 742–751), which the paper itself presents as an improvement over an earlier explicit estimate (Zhou-I, extract l. 1931–1934).

The second paper never names the corollary at all. It inherits the bridge through the first paper, at exactly the bridging proposition (Zhou-II, extract l. 291–292), and it goes further: a remark tracks where one hypothesis is used in the four cited volumes and concludes from that tracking that their results “still hold for” a different class of initial data (Zhou-II, extract l. 267–280). The disputed substep is never touched.

6.2 Silence, and a revealing contrast

Searches over both documents return zero hits for the known critique, for the indeterminacies, for the multiradial construction, for the pilots, and for every term of dispute or correctness we searched for. The positioning is implicit instead, through the word “unconditionally” applied to the consequences (Zhou-I, extract l. 29–30).

What makes this more than an absence is the contrast. Both papers do practice explicit reservations about rigour — the second states plainly that it omits a step and “claim[s] that

we can always do the similar things presented above” (Zhou-II, extract l. 153–156) — but they apply such reservations only to their own omissions, never to the imported step.

6.3 No alternative scale

The log-volume normalization is quoted from the fourth volume of the source theory rather than replaced (Zhou-I, extract l. 474–480). The sharper bounds, the separate readout and the new constants all sit on the sides of the bridge, not in it. One question is expressly left undecided: whether an explicit factor appearing in the imported inequality is a renormalization of the same quantity that the corollary folds into its notation, or a different quantity, could not be decided within the sources of that application, and is recorded as a follow-up rather than answered.

6.4 Result of Case D

Both papers *presuppose* the certificate. Neither circumvents it and neither introduces an alternative scale for it. The first contributes one element of obligation 6 — its own explicit numerical readout on the Θ -side — but supplies no compatibility identity, and it takes obligation 7 over wholesale. The unmet obligation is thus not closed downstream but inherited unexamined, and in the second paper extended to a class of data the cited volumes do not treat.

Searched, not found. Both extracts were searched case-insensitively; the counts are given as first paper / second paper. Scholze 0/0; Stix 0/0; indetermin 0/0; multiradial 0/0; Theta-pilot, q-pilot, theta pilot 0/0; Frobenioid 0/0; controversies, criticis, criticiz, dispute, debate 0/0; gap in, not verif, unverif, validity, correctness 0/0. Two search terms do return hits, and neither is a discussion: the masked 3.12 returns 1/0, the single hit being the citation at (Zhou-I, extract l. 732), and log-theta-lattice returns 1/1, in both cases the title of a bibliography entry (Zhou-I, extract l. 2583–2584), (Zhou-II, extract l. 2254–2255). The reach of this block is the reach of that word list: a discussion conducted under other terminology would not have been caught by it.

Version caveat. The second paper describes itself, in the first line of its abstract, as an older version to be updated (Zhou-II, extract l. 10). Every finding about it is a finding about that version; a later version could in particular argue the inheritance step differently.

7 What the four applications jointly show

7.1 Four different roles, not four detections

It would be convenient, and wrong, to say that the finding has been confirmed four times. The four applications do four different things, and stating them separately is both more accurate and more informative (Table 10).

Only A is a detection. B is the opposite of a second detection: it establishes that a specific candidate site is *not* where the obligation is discharged, which narrows the search rather than repeating it. C is a discrimination case whose value lies in returning a *different* answer. D says nothing about whether the obligation can be discharged; it says that neither of the two versions examined does so.

Case B carries a second function worth naming, because it is the only internal evidence here that the instrument can return something other than a deficit. Under the same criteria that record an unmet obligation in Case A, the text examined in Case B supplies an explicit

Case	Role	What it establishes
A	Detection	Obligations 6 and 7 are not discharged at a named substep: the compatibility identity is not stated, and the order relation is not derived.
B	Closing off a site	The site to which quantification is delegated is explicit and self-contained at the nodes it names, and cannot discharge obligation 7, because it consumes the disputed inequality as an input.
C	Comparative discrimination	The instrument discriminates: three rows deliver, and the unmet obligation sits at a different address, in a different theory, with a different shape — an address revised, in the course of this study, by a closer reading of the same sources.
D	Propagation	Downstream work inherits the unmet obligation without examining it, and in one case extends what is inherited.

Table 10: Systematic roles and distinct methodological functions of the four case applications.

quantified chain and an exactly fitting import at the nodes that carry it: on those questions the answer comes back positive. That is a *positive documentary comparison*, not a validation — Case B is not a site at which obligation 7 could be discharged, so it cannot test the row on which the certificate is most exposed to the charge of being built for its answer.

7.2 The address is stable across independent texts

To provide a complete comparative overview of the empirical findings, Table 11 synthesizes all four applications across their primary sources, theoretical domains, primary BC-Ledger findings, identified gap types, and methodological functions in the argumentative chain.

Case	Primary Source	Theoretical Context	BC-Ledger Finding	Deficit	Role
A	IUT-III, Cor. 3.12	Inter-universal Teichmüller theory	R6a/R6b DOES NOT DELIVER	G (step xi-e)	Detection
B	IUT-IV, §1	Explicit log-volume quantification	R4 PARTLY / Import exact	Z-free	Site closure
C	ATS-IV v2, §6.10, Thm. 6.10.1	Arithmetic Teichmüller spaces	R6 PARTLY (revised)	G (sum-level)	Discrim.
D	Zhou-I/II	Downstream Diophantine apps	R6 INHERITED	G (imported)	Inheritance

Table 11: Synoptic cross-case matrix of all four applications: scope, primary findings, identified gap types, and epistemic roles.

What the four do jointly establish is narrower and, we think, more durable than a count of confirmations. *Two* textually independent bodies of work — the theory of Cases A and B and the program of Case C — examined with one unchanged instrument, place the load of the argument at the same passage: the step from a common container to a derived order relation. Neither completes that step, and neither completes the obligation immediately before it either — though they fail there differently, which is what makes the comparison informative. A third body, Case D, places nothing: it imports the step from the first, as its own text says, and can therefore corroborate nothing — what it adds is the observation that the obligation

travels unexamined. Obligation 6 is unmet in Case A, where the compatibility identity is absent outright; partly met in Case C, where the compatibility holds between the sums and is place-wise disclaimed; and in Case D contributed to on one side only. Obligation 7 is met in none of them in the strict sense of the ledger.

That the instrument nevertheless separates the cases sharply — “does not deliver” versus “partly” versus “inherited” — tells against reading the common finding as a mere artifact of the certificate’s design. It does not settle the question. A genuine positive control — a transfer argument of the same shape whose seventh obligation *is* discharged — was not among the cases examined, and naming that gap is part of the finding rather than an aside to it.

7.3 An independent formalization project reaches the same site

During the same period an international formalization project published an interim report [9] which isolates a compatibility problem at the same passage. Its own words:

“we isolate a specific compatibility problem at the final stage of the argument: the relation between the construction arising directly from the q -pilot in its native arithmetic holomorphic structure and the construction obtained by anabelian and Kummer-theoretic methods through the multiradial procedure. . . . *Project LANA has not yet reconstructed a proof of the required compatibility.*” (LANA, extract l. 11–16)

and, on the status of that statement:

“the proof of the final numerical inequality *hinges on the compatibility* (9-1) *which is not manifestly false. However, we . . . do not have a proof of* (9-1) *at this time.*” (LANA, extract l. 2988–2990)

The required statement is an existence claim: two isomorphisms of pointed real vector spaces with a common target are constructed — one from the q -pilot “together with the native arithmetic holomorphic structure” (LANA, extract l. 2807–2812), the other anabelian and dependent on a choice of integral structure and “subject to indeterminacies” (LANA, extract l. 2783–2803) — and what is wished for is “that *there exists some suitable* S *such that* $\eta^q = \eta_S^{\text{anab}}$ ” (LANA, extract l. 2815–2819).

The report’s R^{ss} quotient identifies adelic measurable sets when their volumes are equal; this definition does not prove equality of the two concrete pilot classes or of the maps in (9-1). The equation remains an open target in the cited report (PDF pp. 46, 48–49). A proposed object-wise role crosswalk follows (Table 12); it does not supply the missing typed implication to obligation 7.

Three statements about this convergence, and deliberately no fourth.

1. **A proposed role correspondence, not validation.** The maps and their requested compatibility can motivate a question at obligation 6. This is not a formal identification with the certificate contract.
2. **The typed bridge remains missing.** An implication from (9-1) to obligation 7 would require the chosen S , the relevant object and quotient representatives, a common normalized log-evaluation, and the orientation and upper bound of the target half-line to be justified. Equality of two abstract maps alone does not supply these premises.
3. **No order conclusion is derived here.** If those premises and an appropriate comparison bound were proved, a compatible evaluation could support an order argument. Neither this paper nor the cited interim report supplies that complete chain. The correspondence is therefore a research task, not a proved one-way implication or a refutation.

Obligation	Object in the report	Anchor
3 — container, typed embeddings	the common target with its two typed maps	2790–2812
4 — permitted identification	stipulated source identifications plus the choice of S	2937–2939
5 — split budget	two indeterminacies on the anabelian side, the third as ambiguity in the choice of S	2786, 2973–2974
6 — compatible evaluation	the common target is by construction the volume evaluation quotient, but the required equality of the two evaluation isomorphisms — a compatibility identity in the sense of this obligation — is demanded and unproved	2790–2794, 2815–2819, 2988–2990
7 — order compatibility	no typed implication established here; object, normalized evaluation and half-line bound remain to be connected	2815–2819

Table 12: Crosswalk between independent formalization report (Project LANA) objects and certificate obligations.

The report’s own position is also worth recording precisely, because it matches the verdict vocabulary of Section 2 rather than either partisan position: the compatibility is “not manifestly false”, and there is no proof of it. That is the same distinction the ledger insists on. *Does not deliver is not refuted* — and the convergence, such as it is, is a convergence on where the question sits, not on an answer to it.

What “independent” means here, and what it does not. The word is used in a strictly documented sense. The certificate and its nine rows were fixed in May 2026, before any of the four applications reported here were carried out; those were carried out in August 2026; the interim report appeared in July 2026 and does not cite this work. That is the whole of the evidence, and it supports exactly one limited chronological observation — that the initial ledger predates the report. This does not rule out influence on later applications or revisions, and it does not establish the absence of contact or influence in either direction, which is a claim about people that no reading of the documents could settle, and which is therefore not made.

Two further limitations belong immediately here. The report is an interim report, its authors record internal disagreement, and our mapping is an object-wise assignment, not a proof of equivalence. The mapping also presupposes rather than exhibits obligations 1 and 2: the two objects and the target half-line are taken as already typed, and only obligations 3 to 7 are matched in the table above.

7.4 What is not shown

1. **Five of nine rows are unfilled.** Rows R0, R1, R3, R5 and R7 were not part of any of the four applications and are open in all of them. R5 was touched only where R4 forced it. A ledger with four rows filled is a partial ledger, and the summary tables above should be read as such. The four are also not of one status, and the denominator should not obscure it: three of them — R2, R4 and R6, the last reported at both halves of its split — are core rows, while R8 is the triggered extension of Section 2.3, which maps to

no obligation and is out of scope where its trigger is absent. A completeness statement about the certificate is a statement about the eight core rows R0–R7, of which three were filled.

2. **No claim about correctness.** Not one proof line was followed in any of the four cases, and no estimate was recomputed. In particular Case B establishes *that* a computation is carried out and where, not that it is right.
3. **No claim about repairability.** Nothing here says the unmet obligation cannot be discharged; one concrete candidate route — whether some text states a theorem deriving the order relation from the declared property pair, rather than inferring it formally — was named as the remaining lever and was not pursued.
4. **No claim about the conjecture.** The word “gap” in this paper means “an obligation of a stated certificate that a quoted wording does not discharge”. It does not mean that the *abc* conjecture is open or closed, and it does not mean that any theory examined here is wrong.

8 Limitations

This section states what the paper does not do, in the terms of the underlying examinations.

1. **No proof was followed.** No theorem of any examined source was verified line by line, no estimate recomputed, no constant checked. What was classified is the structure of justification.
2. **Four rows of nine, three of them core rows.** See Section 7. The row splits of Section 2.4 were not part of the certificate as fixed — two arose from use, the third from the hardened reading of obligation 6 — and they are normative only in the successor instrument [13]. The case reports therefore stay with the original rows wherever the split makes no difference to the finding, so that no application is credited with filling a row that did not yet exist when it was carried out. Where the split does make a difference they use it and say so: Case B reports at R4a and R4b, and Case A at R6a and R6b (Case C’s separation is carried in its running text against the merged row). Version 0.1 stated without qualification that the case reports report against the original rows; that described this paper’s own practice too widely, since Case B already drew its single change of verdict from the R4 split.
3. **One verdict rests on a reconstruction.** In Case A the load-bearing line lost a relation symbol in extraction. The reconstruction and its cross-checks are disclosed in Section 3.2; the general formula “no conclusion depends on the placement of a symbol” would be false for this paper and is deliberately not used.
4. **Case C carries two later corrections in opposite directions** (Section 5.5), and the case must be read with both. The substance examination underlying the second of them is itself a form-and-substance examination, not a refutation; it has appeared as Part 5 [14], as has the source examination underlying the first [15].
5. **Case D describes one version of one document.** The second paper declares itself superseded (Zhou-II, extract l. 10). Its findings are version-bound.
6. **The convergence of Section 7 is object-wise,** not formal. No equivalence between the certificate’s seventh obligation and the report’s compatibility statement is claimed or proved, and the implication we do state runs in one direction only.

7. **Mathematical questions of substance are left open by design**, not by oversight — see Section 2.10, item 2.
8. **The instrument is not validated beyond these applications**. Four applications in two theories are a small sample. Whether the certificate discriminates usefully in other fields is untested; the recipe of Section 2.11 is offered so that it can be tested.
9. **Bibliographic discipline**. Every entry was checked against its original source before this version (verification pass of August 13, 2026). The titles of [7] and [8] were initially omitted — the material underlying this paper recorded their identifiers, author and affiliation but did not quote their titles, and inventing a plausible title would be worse than omitting it — and were subsequently added from the arXiv records themselves. The formalization note of April 2026 that is sometimes mentioned alongside [9] is deliberately *not* cited: it was secured but not read, and an unread source has no place in a bibliography.
10. **Textual basis are extracts**. All quotations come from `pdftotext` extracts; line references are finding aids, not pagination.

9 Conclusion

The disagreement this paper begins from is not, in the first instance, about mathematics. It is about what a particular sequence of sentences *is*. For that kind of disagreement the ordinary instruments are badly suited, because they ask a question — is this correct? — that the two sides answer about different objects, and because they leave a door open to arguments from authority that cannot be closed from inside a prose review.

The bridge certificate replaces that question with one a text can answer about itself: which of seven named obligations does this wording discharge? Around that question sit disciplines that make the answers auditable rather than merely confident — typed rather than graded gaps, an import ledger, recorded search terms for every negative finding, disclosed reconstructions, and a verdict vocabulary in which “not discharged” cannot quietly become “refuted”.

Applied four times, the instrument did four different things, and that is the result worth keeping. It detected two unmet obligations at a named substep and separated them — a compatibility identity that is not stated, and an order relation that is not derived from it — a distinction that matters, because it converts a diffuse suspicion into two specific missing statements instead of one general complaint. It closed off the one site to which the source itself delegates its quantification, by showing that the computation there is explicit, closed at the nodes it names, and downstream of the disputed step. It discriminated when applied to an independent alternative program, returning three delivered rows and a differently shaped gap at a different address — and, when a closer reading of the same sources required it, it revised its own first coding of that case rather than leaving it standing. And it recorded that two downstream applications inherit the unmet obligation in a single sentence without examining it.

What would change this picture is a single, storable object: a theorem that derives the order relation from the permitted identification, rather than inferring it formally from properties attributed to a construction. That such a theorem was not found in the material examined is a statement about that material. Whether it exists elsewhere, or can be proved, is a mathematical question — and it is the question this instrument exists to hand over in a form precise enough to work on.

AI Disclosure / KI-Offenlegung

KI-Systeme wurden in erheblichem Maße in allen Phasen dieser Arbeit eingesetzt: Konzeption und Forschungsdiskussion, Literatur- und Quellenarbeit, mathematische Analyse und Beweisarbeit, Erstellung und Prüfung der Rechenskripte, Ausführung und Auswertung der Berechnungsläufe, Text, Übersetzung und Satz.

Qualitätssicherung: Einsatz mehrerer unabhängiger Modelle verschiedener Anbieter mit Gegenprüfungen und adversarialen KI-Reviews; nachvollziehbare, versionierte Rechenskripte mit gespeicherten Ergebnisdateien und Prüfsummen; fortlaufende Beweis- und Registerdokumente; strikte Claim-Hygiene (Behauptungen nur mit dokumentiertem Beleg, Nachträge statt stiller Korrekturen); Changelogs über alle Versionen.

AI systems were used extensively in all phases of this work: conception and research discussion, literature and source work, mathematical analysis and proof development, creation and checking of the computational scripts, execution and evaluation of the computational runs, text, translation, and typesetting.

Quality assurance: use of several independent models from different providers with counter-checks and adversarial AI reviews; traceable, versioned computational scripts with stored result files and checksums; continuously maintained proof and register documents; strict claim hygiene (assertions only with documented evidence, addenda instead of silent corrections); changelogs across all versions.

Einschränkung für diese Arbeit: Zu dieser Arbeit gehören keine Rechenskripte und keine Berechnungsläufe; die entsprechenden Punkte des Standardtextes (Erstellung und Prüfung der Rechenskripte, Ausführung und Auswertung der Berechnungsläufe, versionierte Skripte mit Ergebnisdateien und Prüfsummen) treffen hier nicht zu. Die Arbeit besteht aus Textbefunden an lokalen PDF-Extrakten.

Restriction for this paper: this work involved no computational scripts and no computational runs; the corresponding items of the standard text (creation and checking of computational scripts, execution and evaluation of computational runs, versioned scripts with stored result files and checksums) do not apply here. The work consists of textual findings against local PDF extracts.

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